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Model Evaluation and Selection

Machine Learning (Bilingual)

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Empirical error

Evaluation Methods

Performance Measure



Empirical error



Error rate is the proportion of the number of misclassified samples compared to the number of all samples.

► If there are m samples and among which a samples have been misclassified, the error rate is $E = a/m$.

Accuracy can be calculated from error rate as $1 - a/m$.



- ▶ Generally, the difference between the output of a learner and the labeled samples is called **error**.
- ▶ The error on the training set of a learner is called the **training error** or **empirical error**.
- ▶ The error on the new or future samples is called the **generalization error**.
- ▶ Obviously, we expect for a learner with small generalization error.
 - ▶ However, as we cannot see future samples, we can only try to minimize the empirical error.



- ▶ **overfitting vs underfitting**
- ▶ There are various approaches to deal with overfitting.
- ▶ However, overfitting cannot be avoided completely.
 - ▶ Machine learning problems are usually NP-hard.
 - ▶ Effective learning algorithms should be finished with polynomial time complexity.
 - ▶ If we can avoid overfitting completely, we proved that $P = NP$ with a constructional method.
 - ▶ If we believe that $P \neq NP$, overfitting cannot be avoided.



Evaluation Methods



Given a set of models $\mathcal{M} = \{M_1, M_2, \dots, M_R\}$, choose a model with smallest generalization error.

The set $\mathcal{M}$ may consist of

- ▶ Instances of same model with **different complexities** or **hyperparameters**. E.g.,
 - ▶ k -Nearest Neighbors: Different choices of k
 - ▶ Polynomial Regression: Polynomials with different degrees
 - ▶ Kernel Methods: Different choices of kernels
 - ▶ Regularized Models: Different choices of the regularization hyperparameter
 - ▶ Architecture of a deep neural network (# of layers, nodes in each layer, activation function, etc)
- ▶ Different types of learning models (e.g., DNN, SVM, KNN, etc.)



- ▶ We can estimate the generalization error based on experiments and then make a choice.
- ▶ To do so, we need a **testing set** to evaluate the discriminant capability over future samples.
- ▶ Then we can use the **testing error** on testing set as an approximation to the generalization error.
- ▶ Generally, we assume that testing samples are i.i.d. sampled from the real distribution of samples.
- ▶ Note that testing set should be mutually exclusive with the training set.
 - ▶ For dataset $\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$, we need to process the dataset to get training set $\mathcal{S}$ and testing set $\mathcal{T}$.



Hold-out split dataset $\mathcal{D}$ directly into two disjoint sets, one as the training set $\mathcal{S}$ and the other as the testing set $\mathcal{T}$.

- ▶ Mathematically, $\mathcal{D} = \mathcal{S} \cup \mathcal{T}, \mathcal{S} \cap \mathcal{T} = \emptyset$
- ▶ After training a model on $\mathcal{S}$, evaluate testing error on $\mathcal{T}$ as the estimation of the generalization error.



- ▶ The partition of training/testing sets should keep the consistency of data.
 - ▶ In classification tasks, the ratio of samples of each class should be kept.
 - ▶ From a sampling perspective, partition keeping the ratio can be obtained by a **stratified sampling** strategy.
- ▶ Even the ratio has been kept, there are multiple ways to make the partition.
 - ▶ Different partition of training/testing sets will give different evaluation results.
 - ▶ When using the hold-out method, multiple random partition and experiments should be conducted, and use the average performance.
- ▶ Fidelity: $2/3 \sim 4/5$ for training and the rest for testing.



Cross validation split the dataset $\mathcal{D}$ into k disjoint subsets,

$$\mathcal{D} = \bigcup_{i=1}^k D_i, \quad D_i \cap D_j = \emptyset (i \neq j),$$

with nearly a same size, each time uses the union of $k - 1$ subsets for training, and the rest subset for testing.



run 1

► There will be k training/testing sets, so can conduct k times training and testing, and return the average of the k testing results.



run 2



run 3



run 4

► Generally called **k -fold cross validation**.
► Often $k = 10$, other choices are 5 and 20.



- ▶ Suppose there are m samples in dataset $\mathcal{D}$, when $k = m$, it is a special case of cross validation called **leave-one-out, LOO**.
- ▶ LOO does not be affected by the partition of dataset as it has only one unique partition.
- ▶ Since the training set is only one sample less than the whole dataset, at most cases, the model evaluated on testing set is very similar to the model trained with the whole dataset.
- ▶ A shortcoming of LOO is that the computational cost of training m models is extremely large when the scale of the dataset is large.



- ▶ In hold-out and cross validation, since part of samples are kept for testing, the training set used to train the model for evaluation is small in size than the dataset $\mathcal{D}$.
- ▶ LOO has an extremely large computational complexity.
- ▶ Based on **bootstrap sampling**, **Bootstrapping** is one good solution.
- ▶ We generate a new dataset $\mathcal{D}'$ by sampling from the given dataset $\mathcal{D}$.
- ▶ Each time, one sample is randomly selected from $\mathcal{D}$, its copy is added to $\mathcal{D}'$, then itself will be put back to $\mathcal{D}$.
- ▶ Such a step will be repeated for m times to get a dataset $\mathcal{D}'$ with m samples.



- ▶ Obviously, some samples in $\mathcal{D}$ will not present in $\mathcal{D}'$, while some other samples may appear multiple times.
- ▶ A simple estimation that the probability of a sample will not be sampled is

$$\lim_{m \rightarrow \infty} \left(1 - \frac{1}{m}\right)^m = \frac{1}{e} \approx 0.368,$$

i.e., after bootstrapping, 36.8% samples in the initial dataset $\mathcal{D}$ will not present in the sampled dataset $\mathcal{D}'$.

- ▶ The left about 1/3 samples can be used for testing purpose. Such results are called **out-of-bag estimate**.



- ▶ In most machine learning algorithms, there are some **parameters** need to be configured. Different configurations will lead to significant differences in performance.
- ▶ In model evaluation and selection, except the selection of learning algorithms, parameters also need to be configured. This is generally called **parameter tuning**.
- ▶ As most parameters are with real values, to train a model for each configuration of parameters is impossible.
 - ▶ In practice, for each parameter, a range and step size are set and evaluated.
 - ▶ If there are 3 parameters, each of them will be evaluated on 5 candidate values, the number of models to be evaluated will be $5^3 = 125$.
- ▶ Note, the data met in practice is generally called **testing data**. To make concepts clear, the testing subset used in model evaluation and selection is called the **validation set**.



Grid Search¹

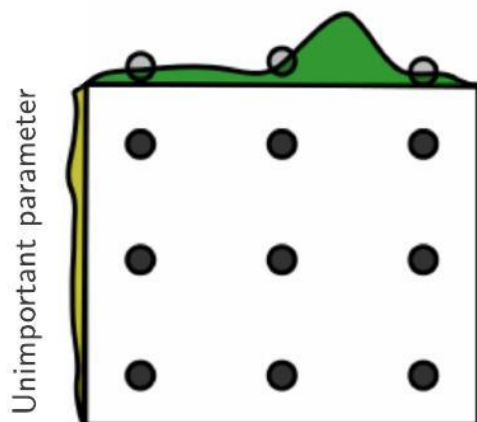
- ▶ Manual optimization gives researchers some degree of insight into goal function;
- ▶ There is no technical overhead or barrier to manual optimization;
- ▶ Grid search is simple to implement and parallelization is trivial;
- ▶ Grid search (with access to a compute cluster) typically finds a better parameter than purely manual sequential optimization (in the same amount of time);
- ▶ Grid search is reliable in low dimensional spaces (e.g., 1-d, 2-d).

¹J. Bergstra and Y. Bengio, “Random Search for Hyper-Parameter Optimization,” **JMLR**, 13: 281–305, Feb. 2012.



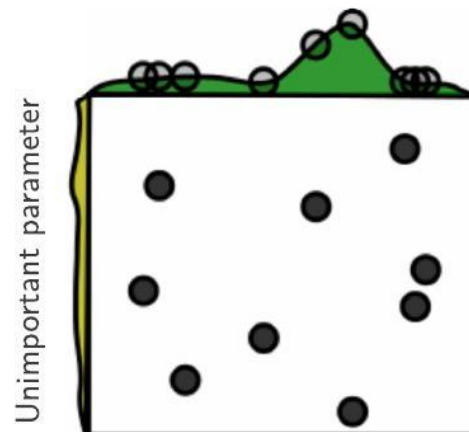
Grid Search vs Random Search²

Grid Layout



Important parameter

Random Layout



Important parameter

- ▶ Random experiments are more efficient because **not all hyper-parameters are equally important to tune.**
- ▶ Grid search experiments allocate too many trials to the exploration of dim. that do not matter and suffer from poor coverage in dim. that are important.

²Bergstra & Bengio, "Random Search for Hyper-Parameter Optimization," **JMLR**, Feb. 2012.



Performance Measure



- ▶ Performance measure reflects the requirements of the learning task.
- ▶ To compare models, different performance measurements will produce different results, which means the good-or-bad of a model is relative.
- ▶ In regression tasks, the frequently used measurement is the **mean squared error**

$$E(f, \mathcal{D}) = \frac{1}{m} \sum_{i=1}^m \left(f(\mathbf{x}_i) - y_i \right)^2.$$

- ▶ More generally, for distribution $\mathcal{D}$ and PDF $p(\cdot)$, the mean squared error can mathematically reformulated as:

$$E(f, \mathcal{D}) = \int_{\mathbf{x} \sim \mathcal{D}} \left(f(\mathbf{x}_i) - y_i \right)^2 p(\mathbf{x}) d\mathbf{x}.$$



- ▶ Error rate and accuracy are the most commonly used measurements for classification tasks. Both can be used in either binary or multiclass classification tasks.
- ▶ Using the indicator function, $\mathbb{I}(i, j) = 1$ if and only if $i = j$, the error rate can be mathematically expressed as

$$E(f, \mathcal{D}) = \frac{1}{m} \sum_{i=1}^m \mathbb{I}(f(\mathbf{x}_i) \neq y_i).$$

- ▶ Correspondingly, accuracy is defined as

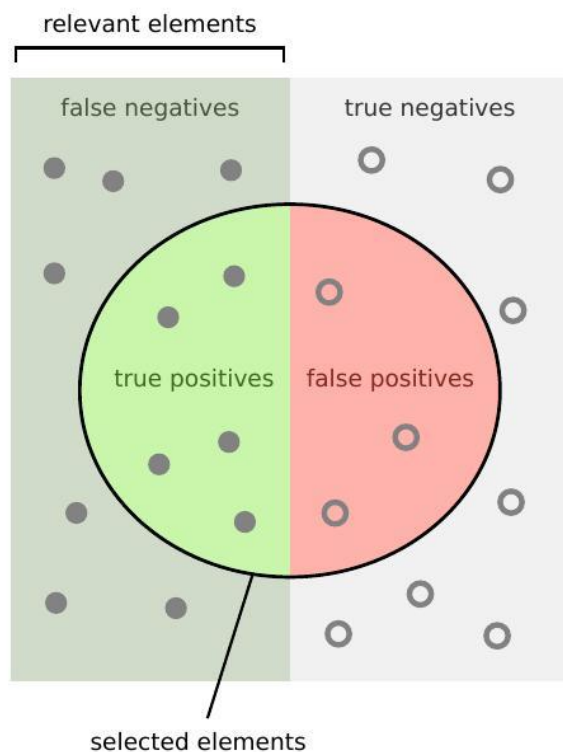
$$\text{acc}(f, \mathcal{D}) = \frac{1}{m} \sum_{i=1}^m \mathbb{I}(f(\mathbf{x}_i) = y_i) = 1 - E(f, \mathcal{D}).$$



- More generally, for data distribution $\mathcal{D}$ and PDF $p(\cdot)$, the error rate and accuracy can be reformulated as:

$$E(f, \mathcal{D}) = \int_{\mathbf{x} \sim \mathcal{D}} \mathbb{I}(f(\mathbf{x}_i) \neq y_i) p(\mathbf{x}) d\mathbf{x},$$

$$\text{acc}(f, \mathcal{D}) = \int_{\mathbf{x} \sim \mathcal{D}} \mathbb{I}(f(\mathbf{x}_i) = y_i) p(\mathbf{x}) d\mathbf{x} = 1 - E(f, \mathcal{D}).$$



How many selected items are relevant?

Precision =



How many relevant items are selected?

Recall =



- ▶ For binary classification tasks, samples can be divided into four classes, which are **true positive**, **false positive**, **true negative**, and **false negative**, based on the compositions of real labels and predicted labels.

- ▶ **Precision** and **recall** are defined as

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN},$$

respectively.

- ▶ Precision and recall are contradicting. If precision is high, recall is generally low. While recall is high, precision will be low.

figure from

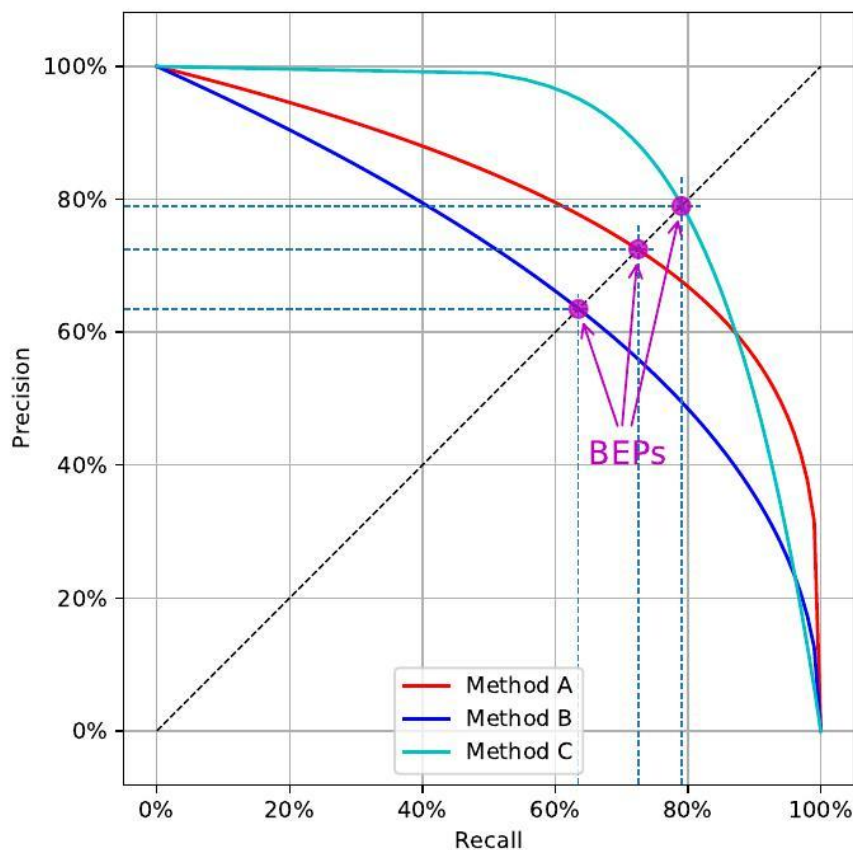
https://en.wikipedia.org/wiki/F1_score



- ▶ In many cases, the samples can be sorted based on the prediction of the learner.
- ▶ At the beginning are the most likely positive samples, while at the end are the most unlikely positive samples.
- ▶ In such an order, each sample is predicted as if it is a positive sample, and corresponding precision and recall rates can be calculated.
- ▶ Taking precision and recall as the vertical and horizontal axes, respectively, the resulting curve is called the **precision-recall curve**, or **P-R curve** for simplicity.



The Precision-Recall Curve



- ▶ If one P-R curve is surrounded by another one completely, then the latter is better than the former.
- ▶ If there is an intersection between two curves, then it is difficult to assert which learner is better.
- ▶ To compare two learner, the area under the curve can be a reasonable criterion.

- ▶ As the area under the P-R curve could be difficult to calculate, some comprehensive measurements are designed.
- ▶ **Break-even point, BEP** is the value that precision equals to recall.



- ▶ A more commonly used criterion is the $F1$ measure,

$$F1 = \frac{2 \times P \times R}{P + R} = \frac{2 \times TP}{m + TP - TN}.$$

- ▶ In some applications, the importance of precision and recall are not the same. There is a more general F_β measure:

$$F_\beta = \frac{(1 + \beta^2) \times P \times R}{(\beta^2 \times P) + R},$$

where $\beta > 0$ specifies the relative importance of recall over precision (Van Rijsbergen, 1979).



- ▶ If there are multiple training/testing, there will be multiple confusion matrix and hence multiple precision and recall rates.
- ▶ To combine these results, we can use the macro- P , macro- R , and the corresponding macro- $F1$, which are defined as:

$$\text{macro-}P = \frac{1}{n} \sum_{i=1}^n P_i, \quad \text{macro-}R = \frac{1}{n} \sum_{i=1}^n R_i,$$
$$\text{macro-}F1 = \frac{2 \times \text{macro-}P \times \text{macro-}R}{\text{macro-}P + \text{macro-}R}.$$



- ▶ Another approach is to take average over all elements in the confusion matrix, and obtain the averages of TP , FP , TN , and FN .
- ▶ These averages are denoted by $\overline{TP}$, $\overline{FP}$, $\overline{TN}$, and $\overline{FN}$.
- ▶ Based on these averages, we can compute the micro- P , micro- R , and the corresponding micro- $F1$:

$$\begin{aligned}\text{micro-}P &= \frac{\overline{TP}}{\overline{TP} + \overline{FP}}, & \text{micro-}R &= \frac{\overline{TP}}{\overline{TP} + \overline{FN}}, \\ \text{micro-}F1 &= \frac{2 \times \text{micro-}P \times \text{micro-}R}{\text{micro-}P + \text{micro-}R}.\end{aligned}$$

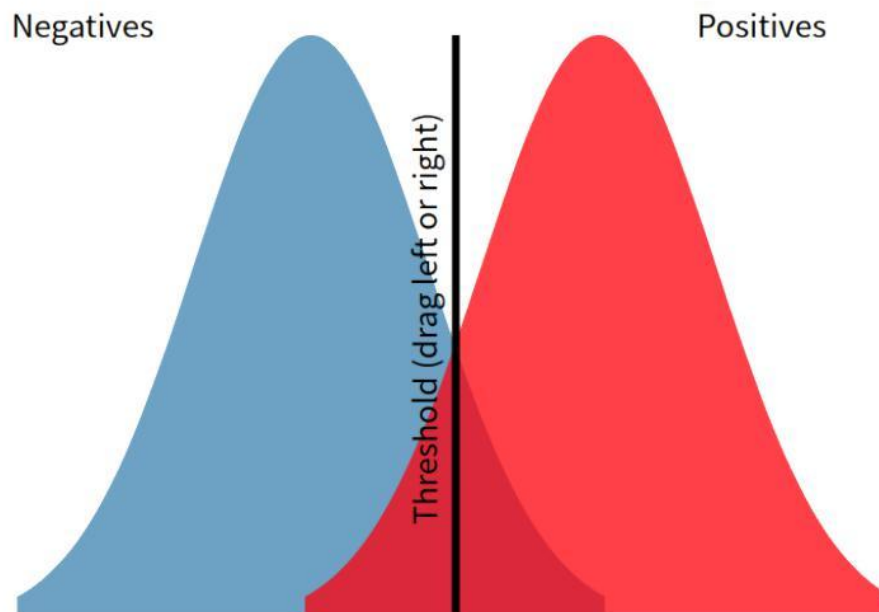
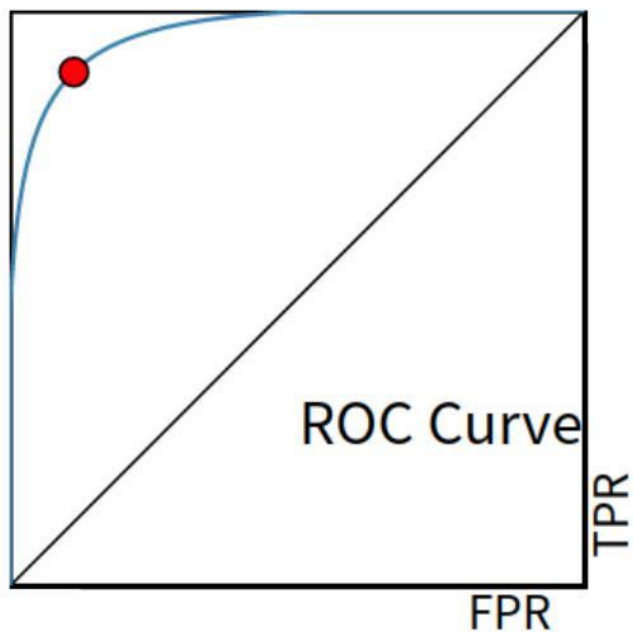


- ▶ Receiver Operating Characteristics, ROC, originated from radar signal analysis technique in detecting planes from the enemy.
- ▶ Similar to the P-R curve, we sort samples based on the prediction of the learner. In this order, each sample is predicted as if it is a positive sample.
- ▶ Different to the P-R curve, the vertical and horizontal axes of the ROC curve are the **true positive rate (TPR)** and **false positive rate (FPR)**, respectively.

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{TN + FP}.$$

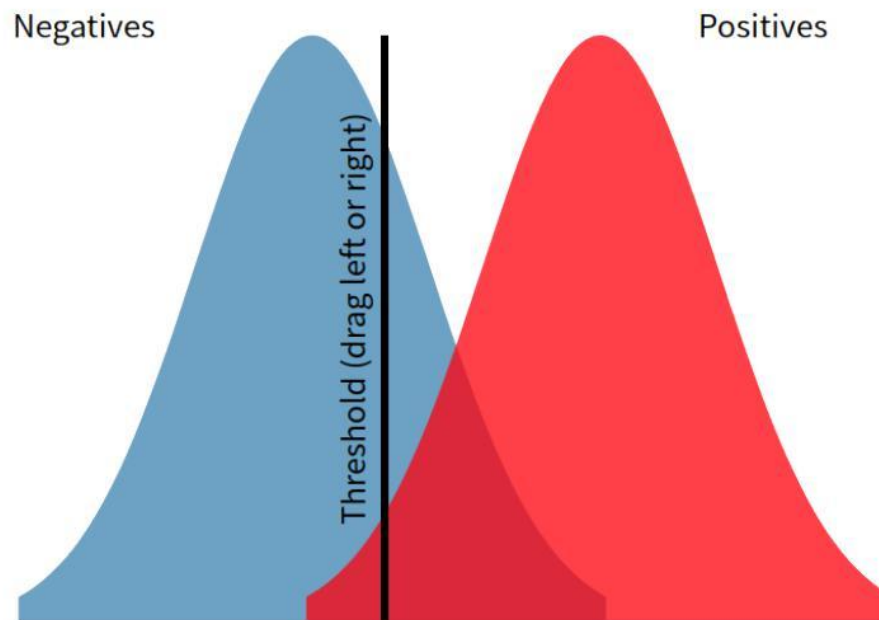
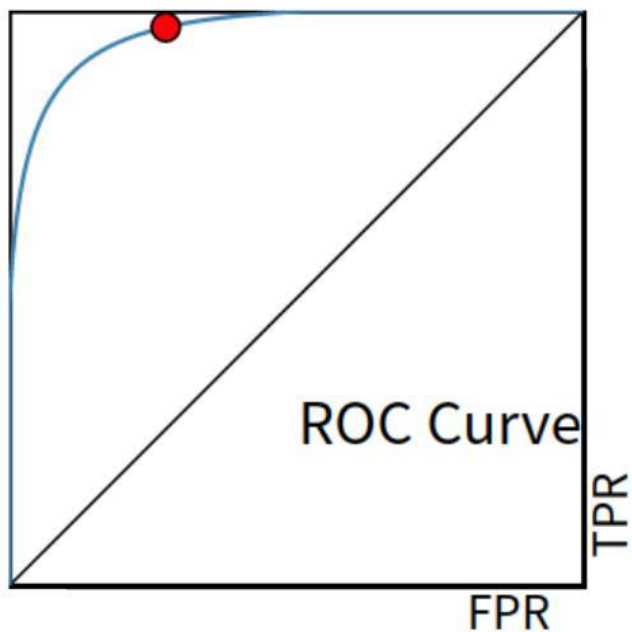


ROC Curve and Bayesian Decision



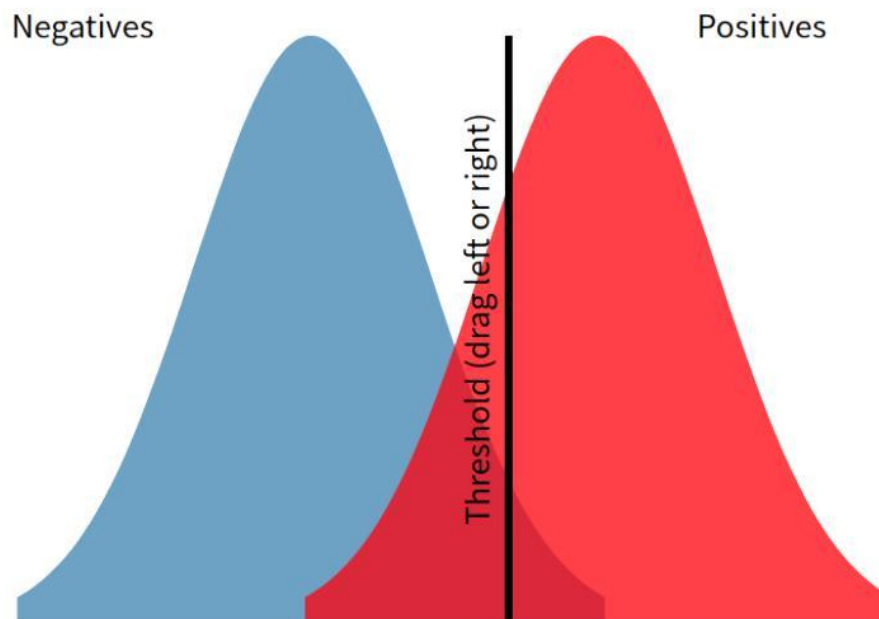
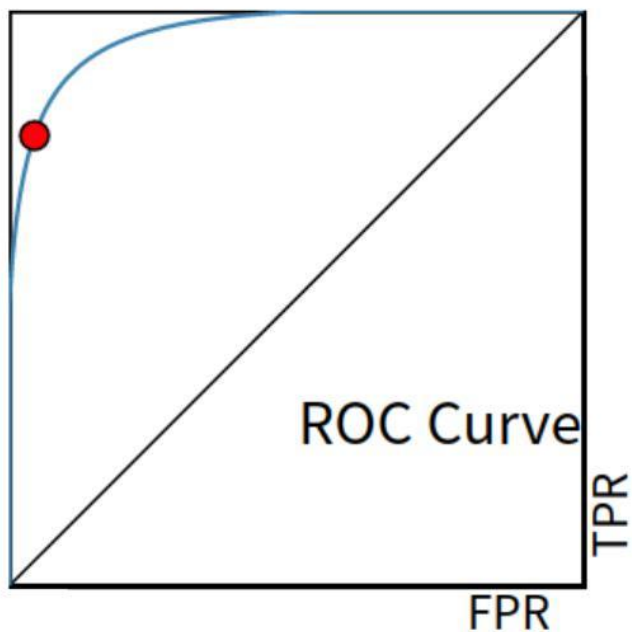


ROC Curve and Bayesian Decision



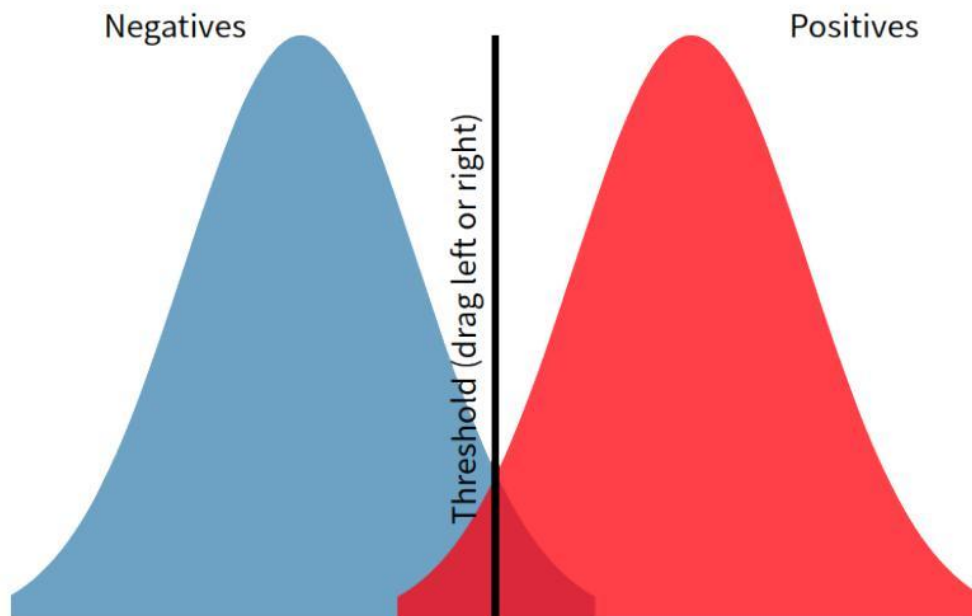
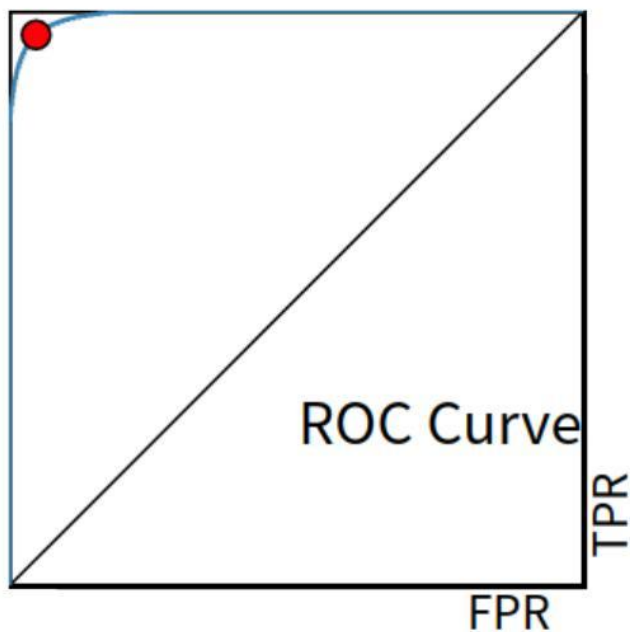


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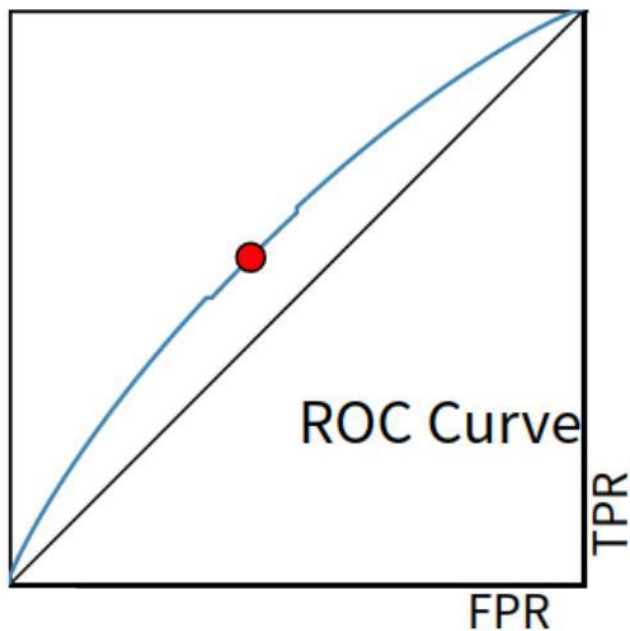


ROC Curve and Bayesian Decision



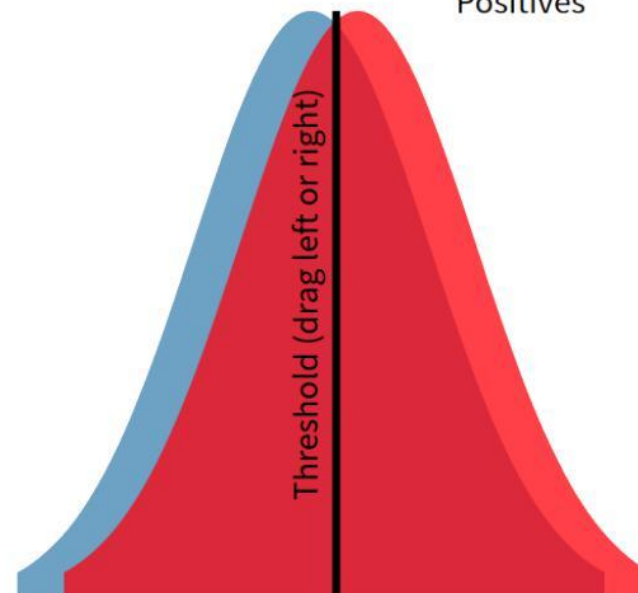


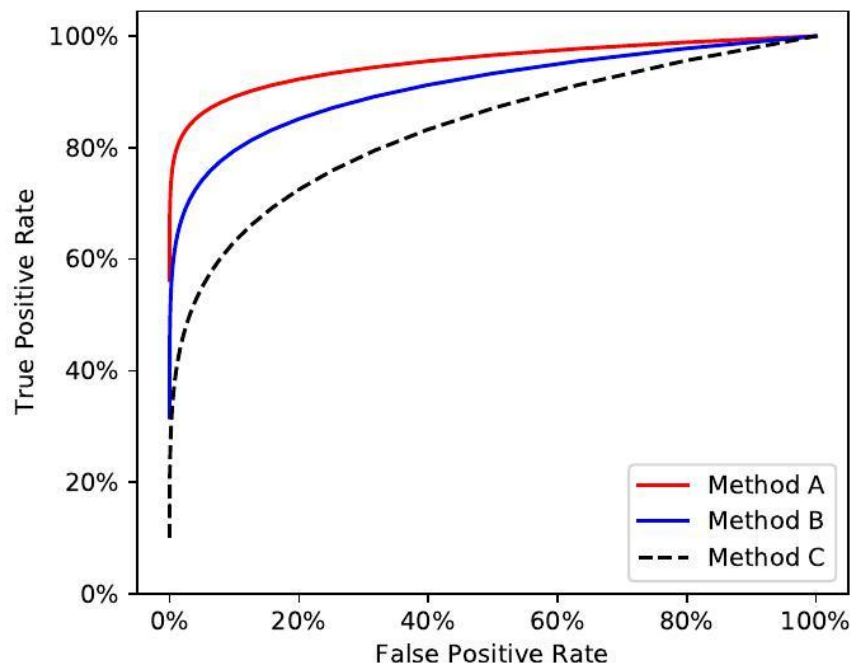
ROC Curve and Bayesian Decision



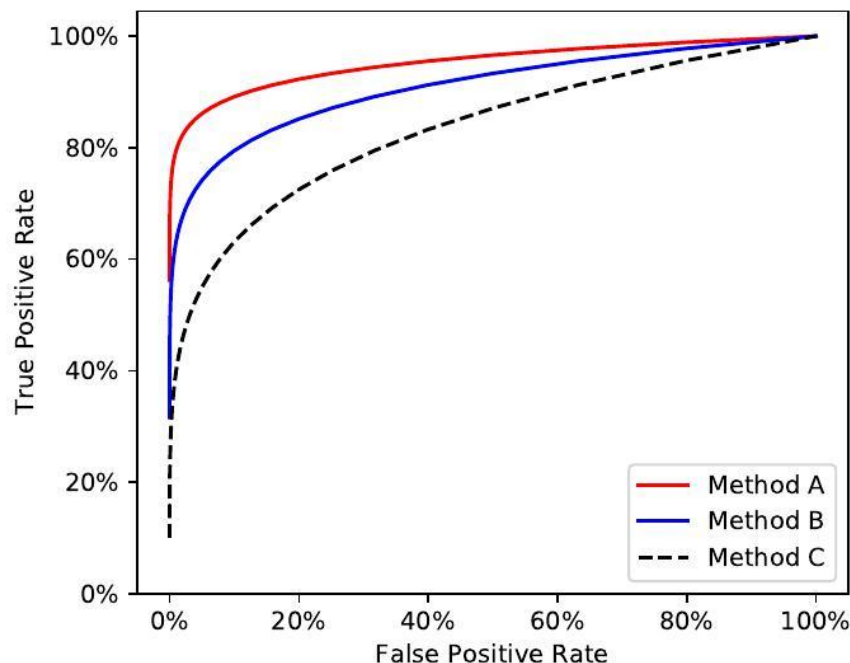
Negatives

Positives





- ▶ Similar to the P-R curve, if one ROC curve is surrounded by another completely, then the latter is better than the former.
- ▶ If there is an intersection between two ROC curves, the Area Under ROC Curve, AUC, is a reasonable criterion.



- ▶ Similar to the P-R curve, if one ROC curve is surrounded by another completely, then the latter is better than the former.
- ▶ If there is an intersection between two ROC curves, the Area Under ROC Curve, AUC, is a reasonable criterion.

- ▶ Suppose the ROC curve is obtained by connecting points $\{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$, then its AUC can be estimated with:

$$\text{AUC} = \frac{1}{2} \sum_{i=1}^{m-1} (x_{i+1} - x_i) \cdot (y_i + y_{i+1}).$$

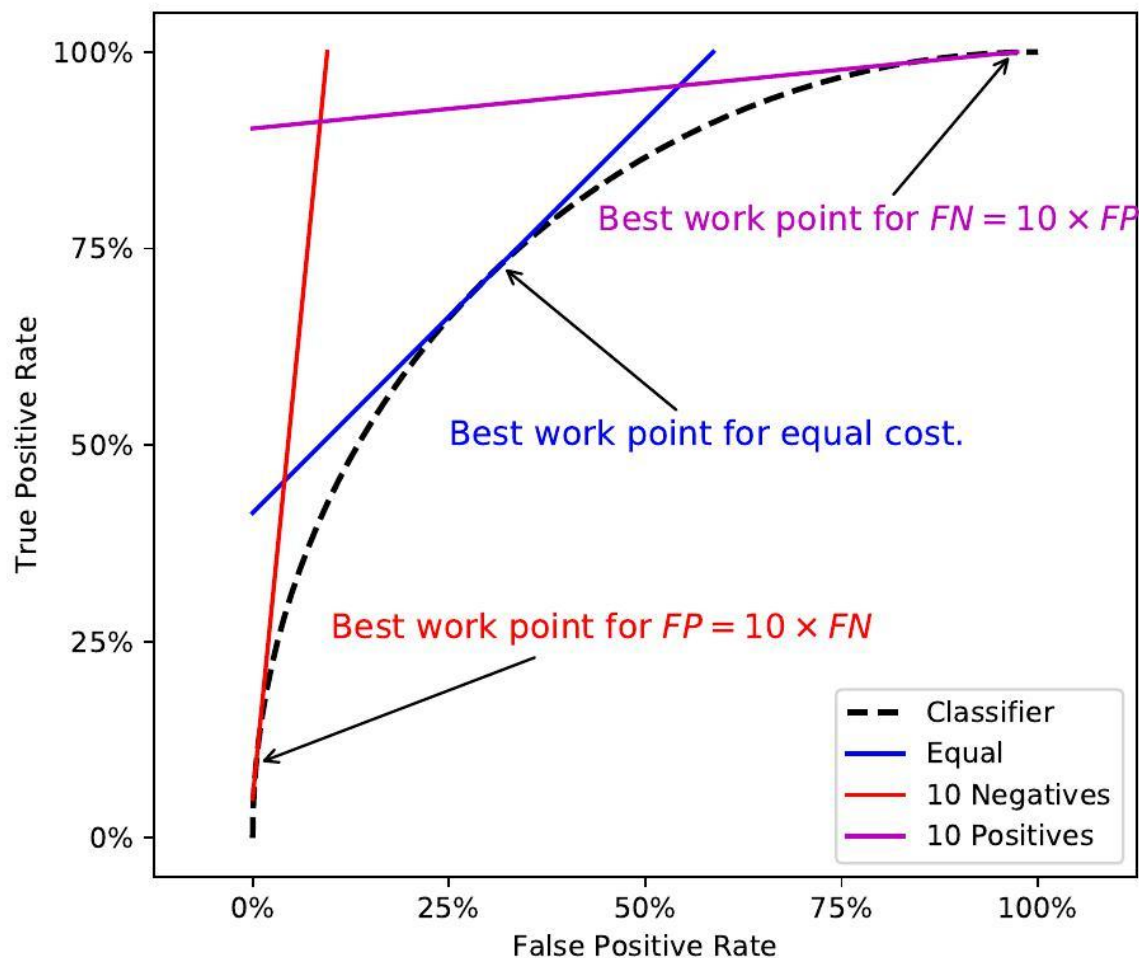


- ▶ AUC considers the quality of the ordering of samples according to the prediction, so it is connected with the ordering error.
- ▶ Given m^+ positive samples and m^- negative samples, and let $\mathcal{D}^+$ and $\mathcal{D}^-$ be the sets of positive and negative samples, respectively.
- ▶ The ordering loss is then defined as:

$$\ell_{\text{rank}} = \frac{1}{m^+ m^-} \sum_{\substack{\mathbf{x}^+ \in \mathcal{D}^+ \\ \mathbf{x}^- \in \mathcal{D}^-}} \left(\mathbb{I}(f(\mathbf{x}^+) < f(\mathbf{x}^-)) + \frac{1}{2} \mathbb{I}(f(\mathbf{x}^+) = f(\mathbf{x}^-)) \right),$$

where for each pair of positive-negative samples, if the positive sample is with a prediction less than the negative one, one point should be penalized; if they are equal, then penalize 0.5.

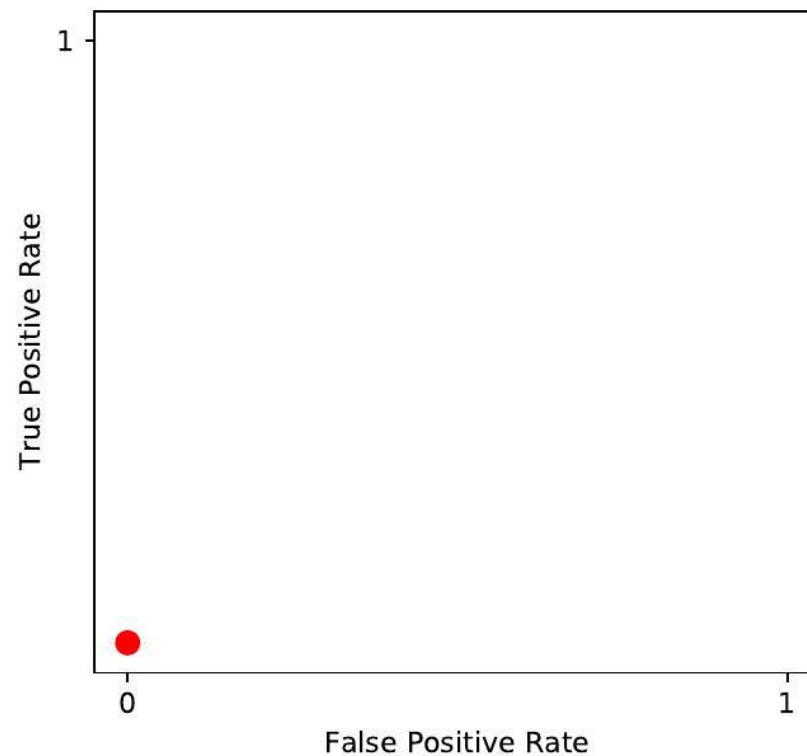
- ▶ Loss ℓ_{rank} corresponds to the area above the ROC curve, so $\text{AUC} = 1 - \ell_{\text{rank}}$.



- Different types of misclassification are associated with different costs.

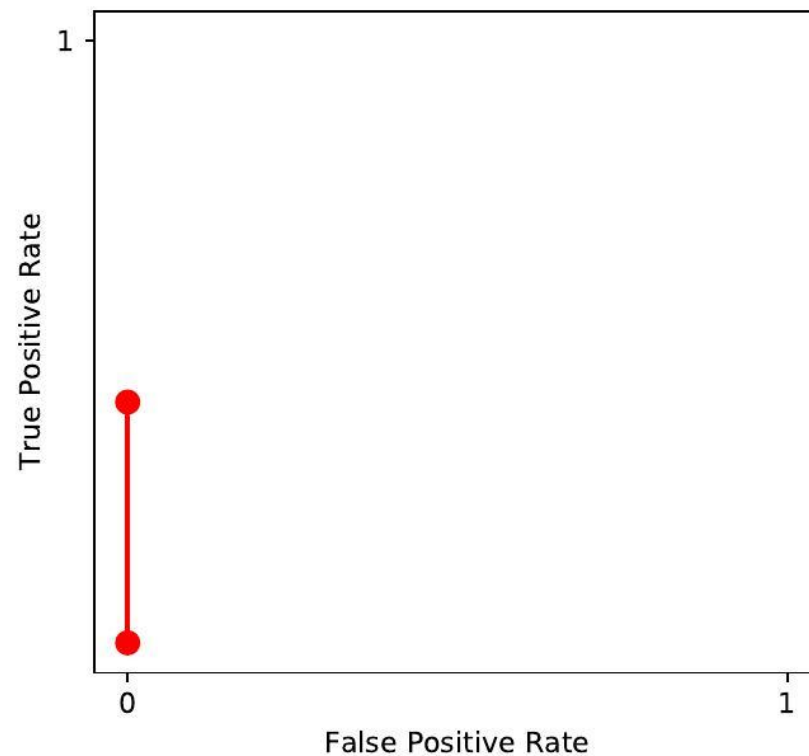


Item	Prob	Class
Ex 9	0.99	+
Ex 7	0.98	+
Ex 1	0.72	-
Ex 2	0.7	+
Ex 6	0.65	+
Ex 10	0.51	-
Ex 3	0.39	-
Ex 5	0.24	+
Ex 4	0.11	-
Ex 8	0.01	-



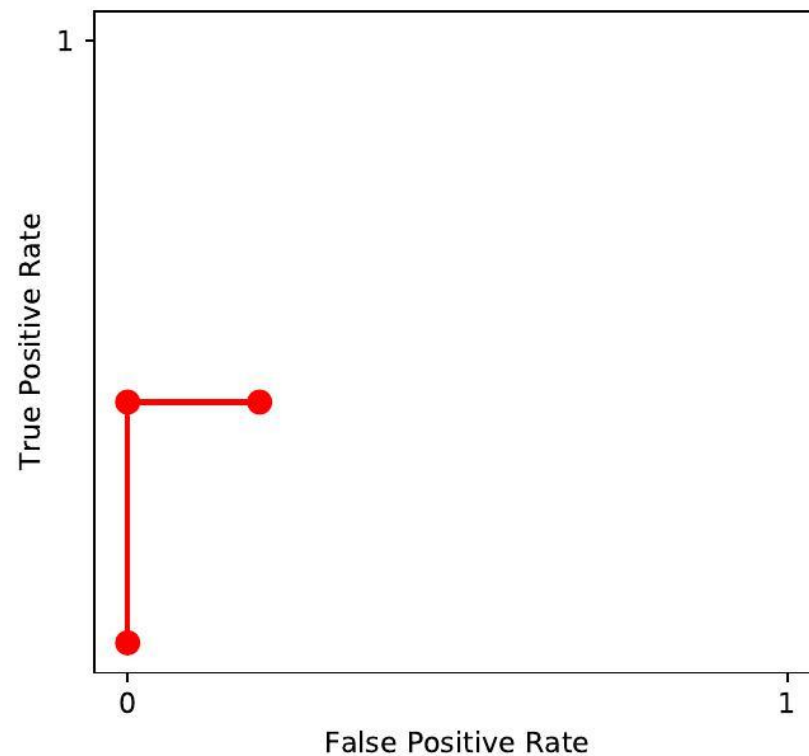


Item	Prob		Class
Ex 9	0.99	TPR=2/5, FPR=0/5	+
Ex 7	0.98		+
Ex 1	0.72		-
Ex 2	0.7		+
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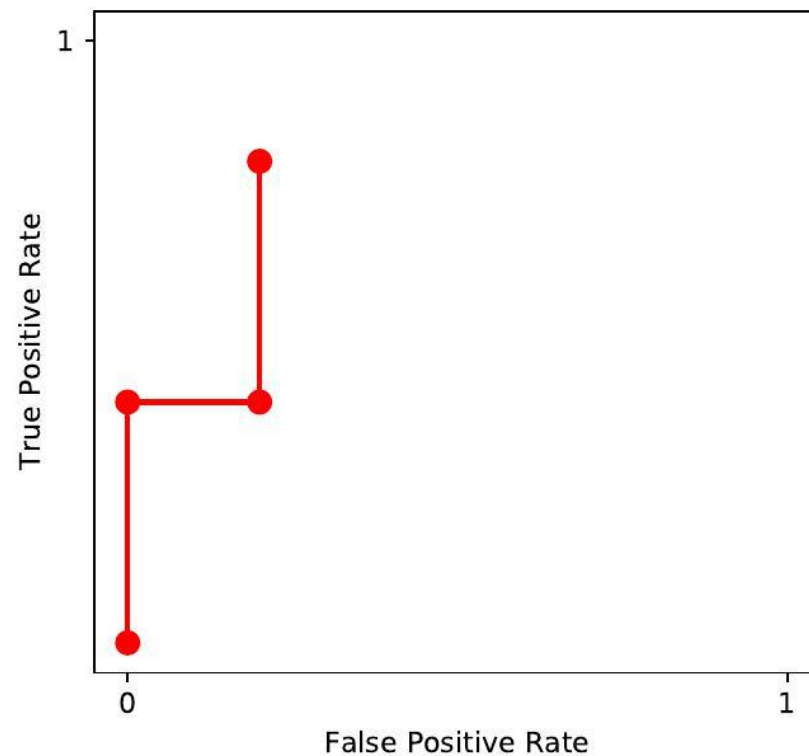


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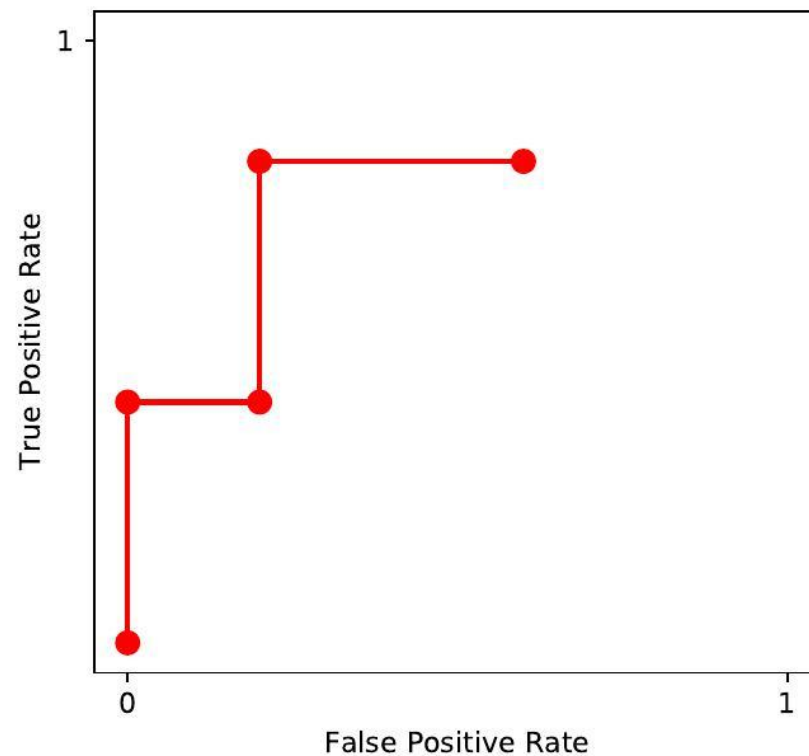


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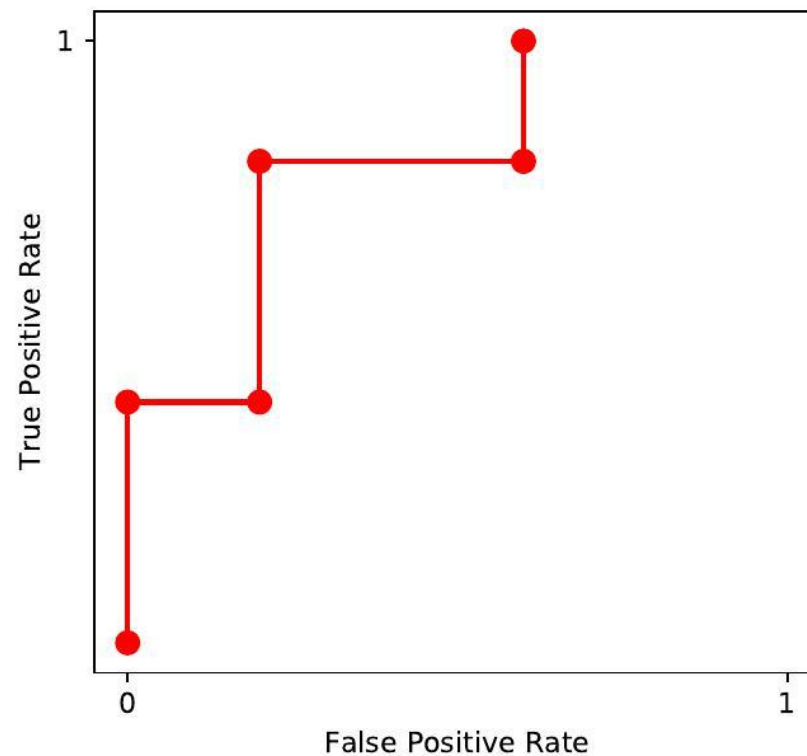


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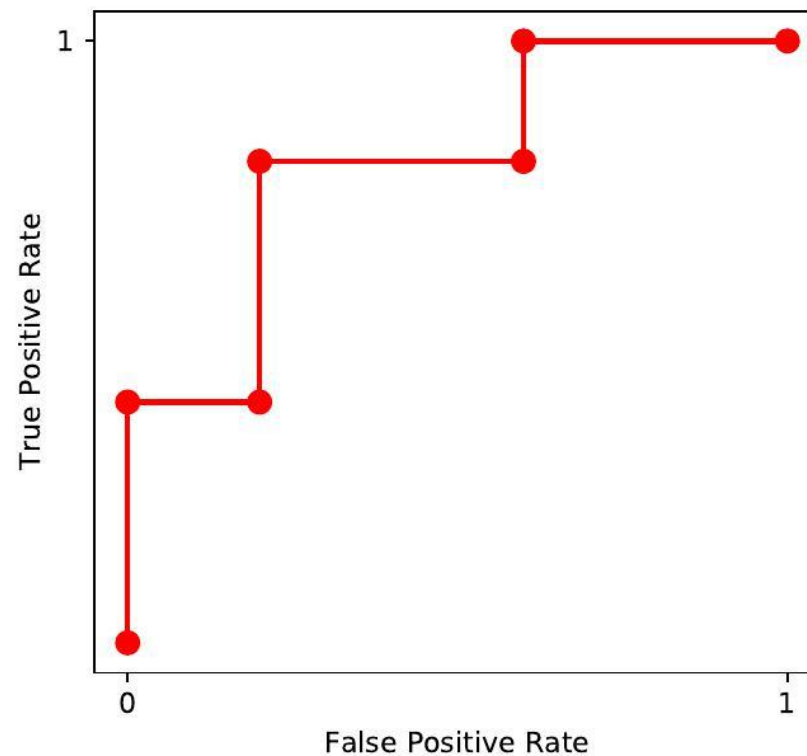


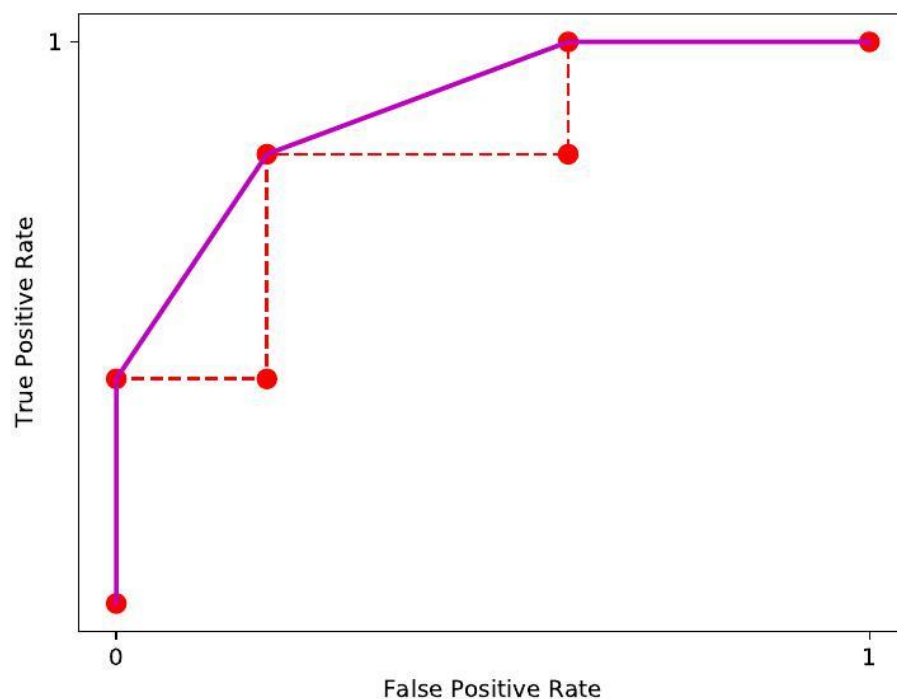
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Ex 5	0.24	TPR=5/5, FPR=3/5	+
Ex 4	0.11		-
Ex 8	0.01	TPR=5/5, FPR=5/5	-





1. Sort samples according to the predicted probability of being a positive sample descending.
2. Loop the sorted samples
 - i. Determine the threshold to classify positive and negative classes
 - ii. Compute TPR and FPR under the above threshold
 - iii. Output the coordinates of (TPR, FPR)